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	(a)	(i)	bodies exert <u>gravitational force</u> on each other <u>provides centripetal force</u> <u>equal in magnitude</u> (and oppositely directed) from <u>Newton's third law</u>	B1 B1
		(ii)	equal centripetal force, so $Mr\omega^2 = m(d-r)\omega^2$ $r = \frac{m}{M+m}d = \frac{1}{\frac{M}{m}+1}d$ $r = \frac{1}{240+1} \times 3.9 \times 10^{11}$ $r = 1.6(2) \times 10^9 \text{ m}$	C1 M1 A1
	(b)		angular velocity = $\frac{2\pi}{2.7 \times 365 \times 24 \times 60 \times 60} (= 7.379 \times 10^{-8}) \text{ rad s}^{-1}$ gravitational force on Arion provides centripetal force $\frac{GMm}{d^2} = m(d-r)\omega^2$ $M = \frac{(3.9 \times 10^{11})^2 \times (3.9 \times 10^{11} - 1.618 \times 10^9) \times (7.379 \times 10^{-8})^2}{6.67 \times 10^{-11}}$ $M = 4.8(2) \times 10^{30} \text{ m}$	C1 M1 A1

4	(a)	direction of acceleration opposite direction of displacement	B1	
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